

LAB 1: Introduction to HTML Basics

Purpose

1. To learn how to inspect code.
2. Understand the working of HTML and its basic tags
3. Browser inspection

Tool:

We are going to use notepad++

Step 1: Open it

Step 2: Write Some HTML

Save the HTML Page

You can use either .htm or .html as file extension. There is no difference, it is up to you.

Step 4: View the HTML Page in Your Browser

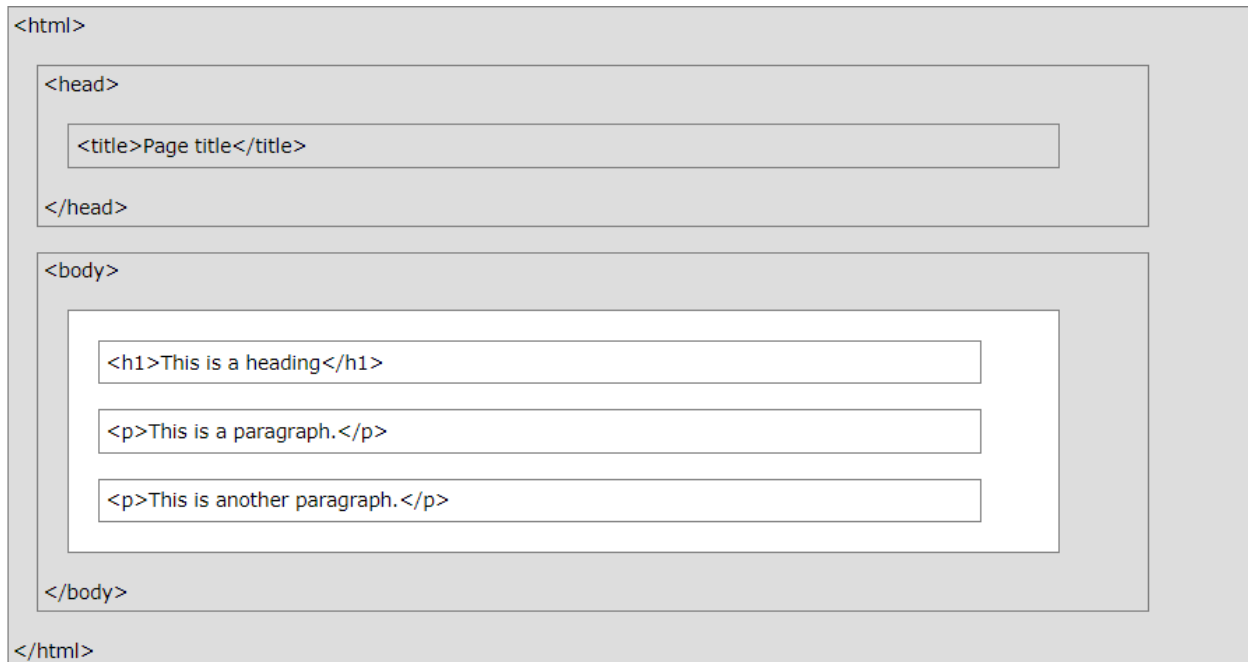
Theoretical Description

1. What is HTML?

HTML is the standard markup language for creating Web pages.

- HTML stands for Hyper Text Markup Language.
- HTML describes the structure of Web pages using markup.
- HTML elements are the building blocks of HTML pages.
- HTML elements are represented by tags.
- HTML tags label pieces of content such as "heading", "paragraph", "table", and so on.
- Browsers do not display the HTML tags but use them to render the content of the page.
- Father of HTML is **Tim Burner Lee**.

HTML Basic structure:



2. HTML Tags:

HTML tags are element names surrounded by angle brackets:

<tagname> content goes here... </tagname>

- **<tagname>** is called as 'Starting Tag'.
- **</tagname>** is called as 'Closing/Ending Tag'.

3. Types of HTML Tags:

Different types of HTML tags are explained in the following:

3.1 HTML Document:

- All HTML documents must start with a document type declaration: **<!DOCTYPE html>**.
- The HTML document itself begins with **<html>** and ends with **</html>**.
- The visible part of the HTML document is between **<body>** and **</body>**.

Example

```
<!DOCTYPE html>
<html>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
</body>
</html>
```

The `<!DOCTYPE>` declaration represents the document type, and helps browsers to display web pages correctly.

It must only appear once, at the top of the page (before any HTML tags).

The `<!DOCTYPE>` declaration is not case sensitive

3.2 HTML Headings:

- HTML headings are defined with the `<h1>` to `<h6>` tags.
- `<h1>` defines the most important heading.
- `<h6>` defines the least important heading:

```
<h1>This is heading 1</h1>
```

```
<h2>This is heading 2</h2>
```

```
<h3>This is heading 3</h3>
```

```
<h4>This is heading 4</h4>
```

```
<h5>This is heading 5</h5>
```

```
<h6>This is heading 6</h6>
```

3.3 HTML Paragraphs:

- HTML paragraphs are defined with the `<p>` tag:

3.4 HTML Links/Anchor Tag:

- HTML links are defined with the `<a>` tag:
- The link's destination is specified in the **href attribute**.

```
<a href="https://www.google.com">This is a link</a>
```

Html link syntax

```
<a href="url">Link text</a>
```

The most important attribute of the `<a>` element is the **href** attribute, which indicates the link's destination.

The *link text* is the part that will be visible to the reader.

Clicking on the link text, will send the reader to the specified URL address.

Example :to create a link to W3Schools.com

```
<a href="https://www.w3schools.com/">Visit W3Schools.com!</a>
```

HTML Links - The target Attribute

By default, the linked page will be displayed in the current browser window. To change this, you must specify another target for the link.

The **target** attribute specifies where to open the linked document.

The **target** attribute can have one of the following values:

- **_self** - Default. Opens the document in the same window/tab as it was clicked
- **_blank** - Opens the document in a new window or tab

Example

Use `target="_blank"` to open the linked document in a new browser window or tab:

```
<a href="https://www.w3schools.com/" target="_blank">Visit W3Schools!</a>
```

HTML Links

1. Use an Image as a Link

To use an image as a link, just put the `<img>` tag inside the `<a>` tag:

Example

```
<a href="https://www.google.com/">  
  
</a>
```

2. Link to an Email Address

Use `mailto:` inside the **href attribute to create a link that opens the user's email program (to let them send a new email):**

Example

```
<a href="mailto:someone@example.com">Send email</a>
```

3. Absolute URLs vs. Relative URLs

```
<h2>Absolute URLs</h2>
<p><a href="https://www.w3.org/">W3C</a></p>
<p><a href="https://www.google.com/">Google</a></p>

<h2>Relative URLs</h2>
<p><a href="link.html">HTML Images</a></p>
```

3.5 HTML Images:

- HTML images are defined with the **** tag.
- The source file (**src**), alternative text (**alt**), **width**, and **height** are provided as attributes.

```

```

```

```

4. HTML Attributes:

Attributes provide additional information about HTML elements.

- All HTML elements e.g <p>, <h> and
 can have **attributes**.
- Attributes provide **additional information** about an element.
- Attributes are always specified in **the start tag**.
- Attributes usually come in name/value pairs like: **name="value"**.

Attribute	Description
Alt	Specifies an alternative text for an image when the image cannot be displayed.
Disabled	Specifies that an input element should be disabled.
Href	Specifies the URL (web address) for a link.
Src	Specifies the URL (web address) for an image.
Title	Specifies extra information about an element (displayed as a tool tip).

5. HTML <head> Element:

- The HTML **<head>** element has nothing to do with HTML headings.

- The **<head>** element is a container for metadata. HTML metadata is data about the HTML document. Metadata is not displayed.
- The **<head>** element is placed between the **<html>** tag and the **<body>** tag:

6. HTML Line Breaks:

- The HTML **
** element defines a **line break**.
- Use **
** if you want a line break (a new line) without starting a new paragraph:

```
<p>This is<br>a paragraph<br>with line breaks.</p>
```

HTML Horizontal Rules

The **<hr>** tag defines a thematic break in an HTML page, and is most often displayed as a horizontal rule.

The **<hr>** element is used to separate content (or define a change) in an HTML page:

```
<h1>This is heading 1</h1>
<p>This is some text.</p>
<hr>
<h2>This is heading 2</h2>
<p>This is some other text.</p>
<hr>
```

How to View HTML Source?

Have you ever seen a Web page and wondered "Hey! How did they do that?"

View HTML Source Code:

Right-click in an HTML page and select "View Page Source" (in Chrome) or "View Source" (in Edge), or similar in other browsers. This will open a window containing the HTML source code of the page.

Inspect an HTML Element:

Right-click on an element (or a blank area), and choose "Inspect" or "Inspect Element" to see what elements are made up of (you will see both the HTML and the CSS). You can also edit the HTML or CSS on-the-fly in the Elements or Styles panel that opens.

LAB ACTIVITY/ CODE

```
<!DOCTYPE html>
<html>
<head>
  <title>My First HTML</title>
</head>
<body>
<p> The HTML head element contains meta data. </p>
<p> Meta data is data about the HTML document. </p>
</body>
</html>
```

LAB TASK

1. Build a simple Web page using Notepad or Notepad++ With a specific title.
Your web page must have:
 - a. An image file
 - b. A description about the image
 - c. Some Description about the title.

2. Open Code Inspection via Developer tools. Experiment around different Web pages and look to it via changing styles.
Note : make 2 to 3 scenarios on customizing web pages and also tell what happens and what are the results

3. **Explain the output of following code with reasoning and also list down the errors in it**

```
<!DOCTYPE html>
<html>
<body>
<h2>HTML Links</h2>
<h3>HTML Links</h3>
<p>HTML links are defined with the a tag:</p>
<p>HTML links are defined with the no tag:</p>
</body>
<a href="https://www.IBM.com">This is IBM Website</a>
```

